A quantitative crossing loss for common conditional noise

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The general crossing identity gives a nonnegative loss, but a weighted comparison needs a quantitative estimate. For the subclass in which the two conditional noise laws coincide, an elementary characteristic-function argument gives a lower bound proportional to the noise variance. A stronger suggestion—a positive gain above the noiseless phase—is false for actual four-point iid laws, including in the moderate-noise range.

1 The common-noise subclass

Let U be centered with support in $[-a_1, a_0]$, where $a_0, a_1 \geq 0$ and $a_0 + a_1 \leq 9/125 = .072$. Let $U_1, \ldots, U_n$ be independent copies, $W = \sum_{i=1}^n U_i$, $s = \mathbb{E}U^2$ and $\lambda = ns$. Here the subscripts label copies, not the two Bernoulli groups: both conditional groups have this same law. For

$$na_0 \leq t < n(1 - a_1)$$

define the unweighted crossing loss by its conditional marginals:

$$g_n(t) = t - \sum_{j=1}^{n} \mathbb{P}(j + W \leq t).$$

This is exactly $t - \mathbb{E}K$ in the general crossing theorem. Its value depends only on the conditional marginals, not on the chosen ordered coupling.

Theorem 1. *Under these hypotheses,*

$$g_n(t) = \mathbb{E}\{t - W\} \;\geq\; \frac{119\lambda}{44(1 + 17\lambda)}. \tag{1}$$

Consequently

$$0 \leq \lambda \leq \frac{9}{25} \quad \Longrightarrow \quad g_n(t) \geq \frac{3}{8}\lambda, \qquad 0 \leq \lambda \leq \frac{8}{25} \quad \Longrightarrow \quad g_n(t) \geq \frac{2}{5}\lambda.$$

The result holds for every positive integer n and includes threshold atoms with the closed-CDF convention.

Proof. Use the auxiliary coupling $\widetilde{T}_j = j + W$. Every conditional marginal is correct. The endpoint bounds imply $0 \leq t - W < n$ surely, so its crossing index is $\lfloor t - W \rfloor$. Hence

$$g_n(t) = t - \mathbb{E}\lfloor t - W \rfloor = \mathbb{E}\{t - W\},$$

using $\mathbb{E}W = 0$. The floor includes its integer endpoint, as required for the closed lower crossing.

Put $\phi = \mathbb{E}e^{2\pi \mathrm{i} U}$ and let U' be an independent copy. The elementary cosine inequality $1 - \cos z \geq z^2/2 - z^4/24$ gives, for $|x| \leq 9/125$,

$$1 - \cos(2\pi x) \geq \left(2\pi^2 - \frac{2}{3}\pi^4 (9/125)^2\right) x^2 \geq 17x^2.$$

The last inequality follows from $3 < \pi < 22/7$ and the rational check

$$18 - \frac{2}{3}(22/7)^4(9/125)^2 > 17. \tag{2}$$

Thus

$$1 - |\phi|^2 = \mathbb{E}[1 - \cos(2\pi(U - U'))] \geq 34s.$$

Centering and $(U + a_1)(a_0 - U) \geq 0$ give $s \leq a_0 a_1 \leq (9/125)^2/4$, so $34s < 1$. For $r = n/2$, convexity of $(1 - x)^{-r}$ on $[0, 1)$ yields

$$|\phi|^n \leq (1 - 34s)^{n/2} \leq \frac{1}{1 + 17ns}.$$

For $0 \leq x \leq 1$, $\sin^2(\pi x) \leq \sin(\pi x) \leq \pi x$. Independence of the copies of U therefore implies

$$\begin{aligned} g_n(t) &\geq \frac{1}{\pi}\mathbb{E}\sin^2(\pi(t - W)) \\ &= \frac{1 - \Re\{e^{2\pi i t}\overline{\phi}^n\}}{2\pi} \\ &\geq \frac{1 - |\phi|^n}{2\pi} \geq \frac{119\lambda}{44(1 + 17\lambda)}. \end{aligned}$$

At $\lambda = 9/25$ the coefficient of λ in the last expression is $2975/7832 > 3/8$; at $\lambda = 8/25$ it is $2975/7084 > 2/5$. Its monotonicity proves the stated linear bounds. □

For completeness, the two numerical bounds on π need no numerical library. Using $\pi = 4\int_0^1 (1 + x^2)^{-1}\,dx$, the finite geometric lower polynomial through $-x^{14}$ has integral greater than $3/4$. Also polynomial division gives

$$\frac{22}{7} - \pi = \int_0^1 \frac{x^4(1 - x)^4}{1 + x^2}\,dx > 0.$$

The cosine bound follows by integrating the fourth-derivative bound $\cos z \leq 1$ in Taylor's formula.

Relation to the acquired strips. If the conditional laws coincide, the lower bound for the second group and the upper bound for the first force their common law into $[-a_1, a_0]$. Thus its diameter is at most the inward sum, even though the separate outer amplitudes can be .6 or .7. The theorem applies throughout the right noise ceiling $\lambda < .36$ and the left ceiling $\lambda < .32$. For $p \in [.25, .53]$, $n \geq 42$, $|H| \leq 1.84$ and $t = np + H\sqrt{p(1 - p) + s}$, the interior-threshold condition follows as in the retained crossing theorem. Indeed $\sqrt{1/4 + .36/42} < .515$, and the displacement from np is less than .948, smaller than either endpoint margin already at $n = 42$. These are conditional range checks, not a proof of acquisition.

2 A gain above the noiseless phase is impossible

Write $u = \{t\}$. In the absence of noise, $g_n(t) = u$. Even in the common-noise subclass, a positive bound for $g_n(t) - u$ cannot hold.

Proposition 2. *Suppose additionally that U is symmetric. If $1/2 \leq u < 1$, then*

$$g_n(t) \leq u.$$

For $U = \pm 1/100$ with equal probabilities, $n = 42$, $p = 1/2$ and $t = 21 + 3/4$, the inequality is strict:

$$g_{42}(t) - \frac{3}{4} = -2^{-42} \sum_{m=0}^{8} \binom{42}{m} < 0. \tag{3}$$

This is an actual four-point iid mixture satisfying the right and left noise-support caps and the raw-threshold bound.

Proof. For each real w and $u \in [1/2, 1)$,

$$\lfloor u - w \rfloor + \lfloor u + w \rfloor \geq \lfloor 2u \rfloor - 1 \geq 0.$$

Symmetry therefore gives $\mathbb{E}\lfloor u - W \rfloor \geq 0$, while centering gives $g_n(t) - u = -\mathbb{E}\lfloor u - W \rfloor$.

In the explicit example, $W = (2M - 42)/100$ with $M \sim \mathrm{Bin}(42, 1/2)$ and $|W| \leq .42$. The floor $\lfloor 3/4 - W \rfloor$ equals one exactly when $M \leq 8$, and otherwise zero. This proves (3). Its noise is $\lambda = 42/10000 = 21/5000$. The standardized raw threshold is

$$H = \frac{75}{\sqrt{2501}} < 1.84.$$

The actual marginal is $Y = B + U$, with independent $B \sim \mathrm{Bern}(1/2)$ and $U = \pm .01$, and $X = (Y - 1/2)/\sqrt{.25 + .0001}$. It is a standardized four-point iid law, not a moment relaxation. □

The failure is not confined to tiny noise. With $U = \pm 1/40$, $n = 512$, $p = 1/2$ and $t = 256 + 3/4$, one has $\lambda = 8/25 = .32$ and $H^2 = 900/401 < 1.84^2$. Symmetry gives nonpositive excess; it is strictly negative because $W = 3/10$ has positive probability and

$$\lfloor 3/4 - 3/10 \rfloor + \lfloor 3/4 + 3/10 \rfloor = 1.$$

A left-strip example is $n = 160$, $p = 7/20$, the same common noise and $t = 56 + 3/4$: here $\lambda = 1/10$, inward sum .05, and $H^2 = 900/365 < 1.84^2$. Both examples satisfy the corresponding outer amplitude bounds as well. They are not asserted to satisfy extremizer contact; their role is to test a proposed implication from the stated noise/threshold hypotheses.

There can also be exactly zero excess at positive noise: in the 42-summand example, choose $t = 21 + 1/2$. Since $|W| \leq .42$, $\lfloor 1/2 - W \rfloor = 0$ surely, and $g_n(t) = 1/2$.

3 Limits of this result

The proof of (1) uses one common noise law and independence of its copies. It does not establish the same constant for unequal conditional laws. Independence is essential: if all 42 noises equal one common random sign times $1/42$, then $W = \pm 1$. At $t = 21$ the loss is zero, although the sum of the marginal variances is $1/42 > 0$ and the support/threshold caps hold. This is a counterexample to dropping independence, not an iid counterexample.

The bound is unweighted. The full comparison contains $\sum_i \pi_i g_i(t)$, signed binomial curvature, phase interpolation, and normal and third-moment repairs. A lower bound for $\sum_i g_i(t)$ alone does not allocate this loss to indices with sufficiently large π_i. Accordingly no retained comparison certificate is removed here. The exact checker independently reconstructs the common-noise sums and the conditional count sums, including all closed collisions used above.